Hybrid Neighborhood Evaluation in Tabu Search and Simulated Annealing for the Permutation Flow Shop Problem

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Outline

Introduction

Permutation Flow Shop Problem (PFSP) — n jobs, m machines, same order, minimize $C_{\max}$. NP-hard for $m \geq 3$.

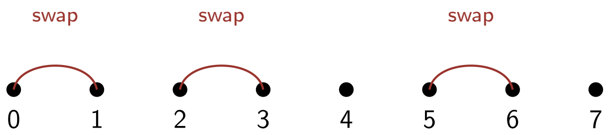
We compare three neighborhoods inside Tabu Search and Simulated Annealing:

Adjacent → ***Fibonacci*** → **Dynasearch**

Adjacent	***Fibonacci***	**Dynasearch**
single swap	disjoint set of swaps	recursive sequence
cheap, shallow	balanced	deep, expensive

Composite Disjoint-Adjacent (*Fibonacci*)

Subset $S \subseteq \{1, \ldots, n-1\}$ of **non-overlapping** adjacent-swap positions; all selected swaps applied simultaneously. Number of valid subsets $= F_{n+1}$ — hence the nickname.



Optimal subset (minimal $\sum_i \Delta_i$) chosen by DP in $O(n)$ given the precomputed $n-1$ swap deltas.

Cost Comparison

Neighborhood	Candidates	Total per iteration
Adjacent	$n-1$ swaps	$O(mn)$
Fibonacci	$n-1$ swaps + DP	$O(mn^2)$
Dynasearch (rec.)	$O(n^2)$ segments	$O(mn^3)$

Fibonacci explores an **exponentially-sized** neighborhood (F_{n+1} valid subsets) at **quadratic** cost — the sweet spot between cheap Adjacent and expensive Dynasearch.

Tabu Search & Simulated Annealing

Tabu Search

- Tabu list keyed by move or composite signature.
- Tenure $\tau_{\text{tabu}} = 10$.
- Aspiration on global best improvement.

Simulated Annealing

- Multiplicative cooling $T \leftarrow \alpha T$.
- **Time-based reheating** on stagnation ($\tau_{\mathrm{stagn}} = 500$ ms).
- Standard Metropolis acceptance.

Same neighborhoods $\Rightarrow$ performance differences attributable to neighborhood design, not framework.

Experimental Setup

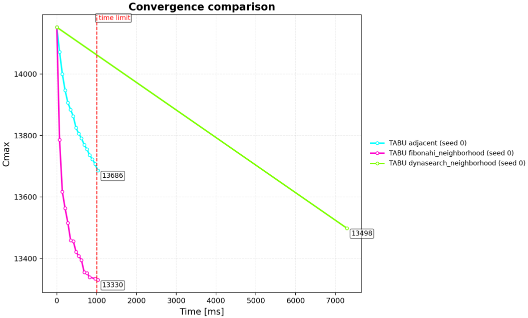
- Taillard instances, $n \in \{20, 50, 100, 200, 500\}$, $m \in \{5, 10, 20\}$ — 12 instance classes.
- Time limits: $tl \in \{100, 500, 1000, 2000, 5000, 10000\}$ ms.
- SA: $T_0 = 1000$, $T_{\mathrm{final}} = 1$, $\alpha = 0.95$, $r = 1.5$, stagnation 500 ms.
- Metric: PRD to best known $C^*_{\max}$ in %.
- MacBook Pro M4 Pro, 24 GB RAM, Python 3.11.

Results — PRD at $tl = 5{,}000$ ms (selected)

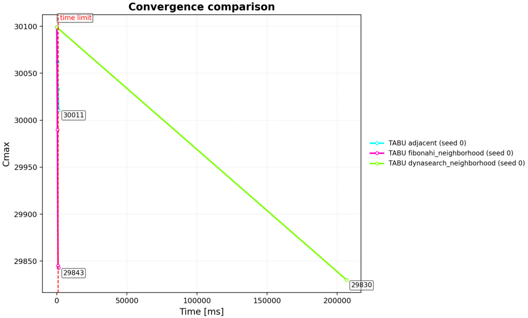
instance	n	m	$\mathrm{fib}_{\mathrm{TS}}$	$\mathrm{fib}_{\mathrm{SA}}$	$\mathrm{dyn}_{\mathrm{TS}}$	$\mathrm{dyn}_{\mathrm{SA}}$	$\mathrm{adj}_{\mathrm{TS}}$	$\mathrm{adj}_{\mathrm{SA}}$
tai20_5	20	5	17.39	17.39	6.18	6.18	**4.17**	6.24
tai50_10	50	10	20.23	20.23	**6.95**	6.95	17.10	9.72
tai100_10	100	10	17.19	17.19	**6.57**	6.48	16.52	8.27
tai200_10	200	10	**12.71**	12.71	13.20	13.20	13.19	8.53
tai200_20	200	20	**18.60**	18.60	20.71	20.71	19.83	20.20
tai500_20	500	20	13.91	13.90	14.51	14.51	**15.34**	15.82

Crossover at $n = 200$: *Fibonacci* **overtakes** both Adjacent and Dynasearch.

Convergence — $tl = 1{,}000$ ms



tai200_20, $n = 200$, $m = 20$



tai500_20, $n = 500$, $m = 20$

Fibonacci (magenta) drops $C_{\max}$ fastest in the first ~ 100 ms; Dynasearch (orange) barely finishes one iteration within tl.

Conclusion and Future Work

Summary:

- Three neighborhoods compared inside TS and SA on Taillard.
- *Fibonacci* (Composite Disjoint-Adjacent) dominates from $n = 200$ onward.
- Dynasearch reaches deeper optima but is impractical for $n \geq 200$.
- SA's time-based reheating helps escape composite-move plateaus.

Future work:

- Incremental delta evaluation for the composite move.
- Adaptive switching between neighborhoods at runtime.
- Comparison with state-of-the-art FSP solvers (iterated LS, VND).